CUTE Control Systems and Data

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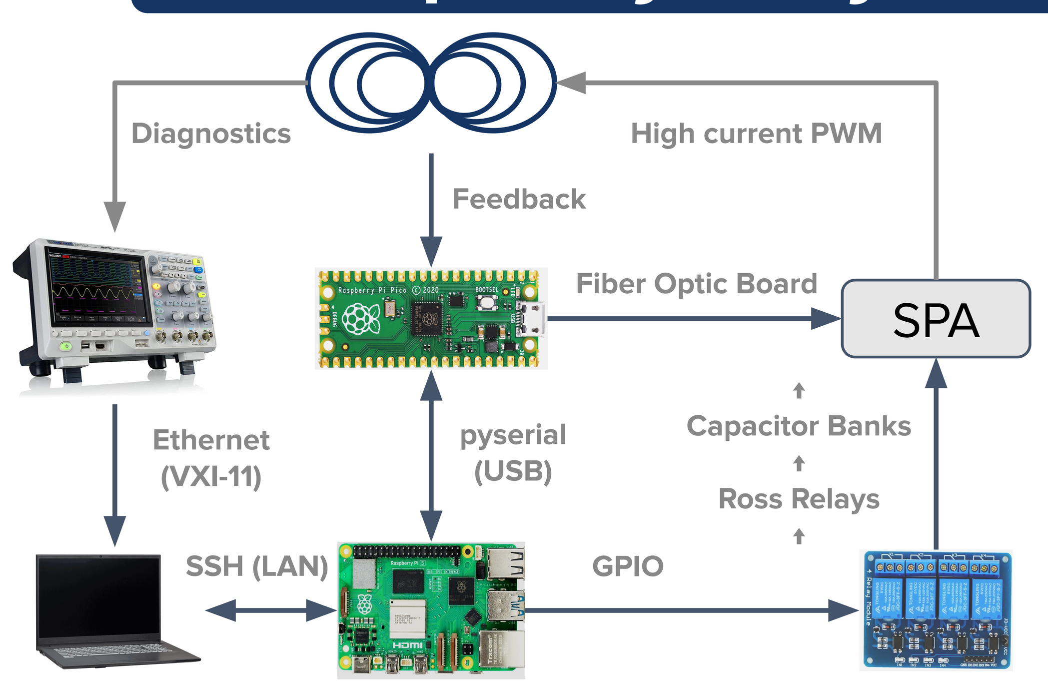
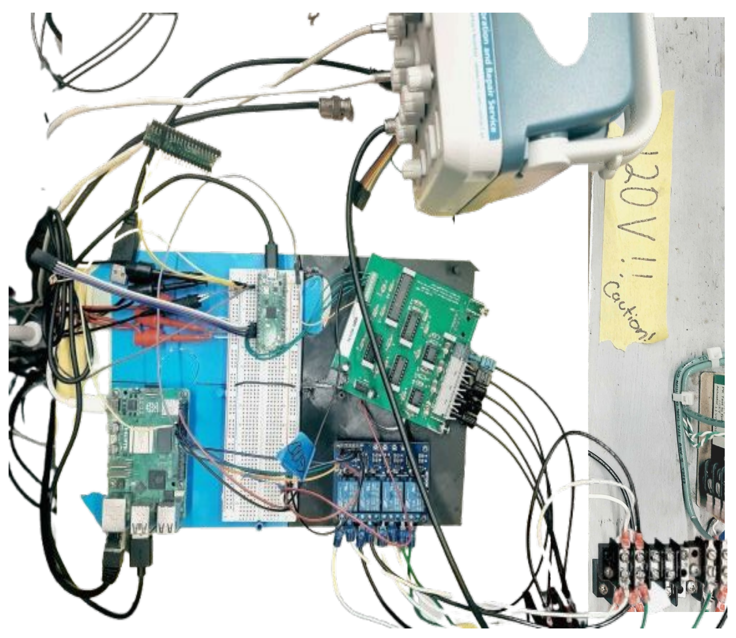
Motivation

- Columbia University Tokamak for Education (CUTE) utilizes multiple capacitor banks that deliver high current pulses to poloidal and toroidal field coils.
- Control system is developed to coordinating charging and dumping, output desired waveforms, and enforce safety.
- This project designs and tests the real-time dual-controller logic diagnostics, and data acquisition via low voltage setup.

Raspberry Pi System Controller



System architecture

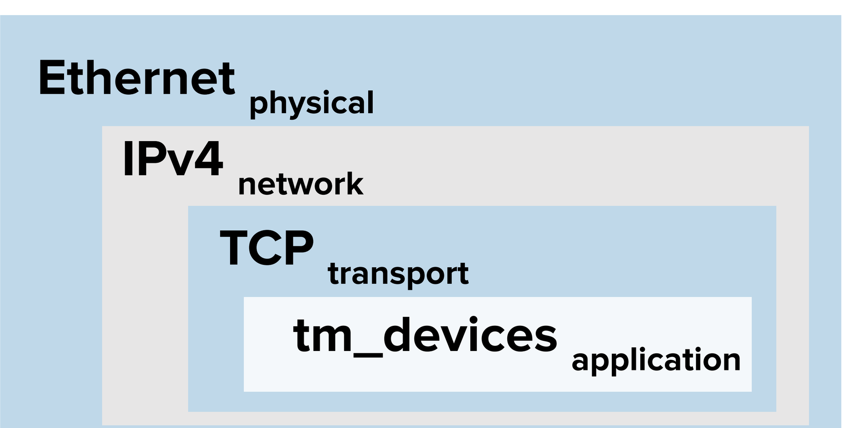


Testing setup

- The system consists of a Raspberry Pi 5 managing user I/O and safety sequences and a Raspberry Pi Pico micro-controller handling microsecond-level gate driving.
- Raspberry Pi 5 manages HV access switch, power supply charging and dumping relays via GPIO outputs, processes user waveform inputs and sends shot profiles over a custom USB serial protocol to the Pico.
- Raspberry Pi Pico triggered by Pi 5 generates 50 kHz PWM pulses to drive the SPA's H-bridge IGBTs.
- Digital outputs from the Pico are converted to optical signals and transmitted via fiber-optic cables to the SPA gate drivers.

Data acquisition and processing

- The Tektronix DPO 2014B Digital Phosphor Oscilloscope acquires the data from Pearson probes and capacitor banks, sending it through the VXI-11 standard over TCP/IP to the computer.
- It is queried by the computer using tm-devices package and PyVISA backend through a static Ethernet Link.



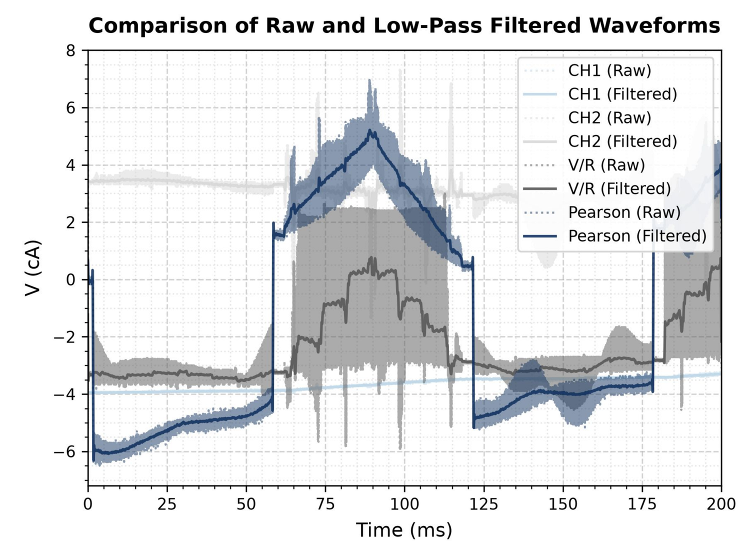
Open Systems Interconnection model

- PC: establish communication, trigger
- Oscilloscope: probe voltage
- Oscilloscope: binary waveform
- PC: scaling, filtering, plotting data

Data acquisition pipeline

Low Voltage Testing Setup

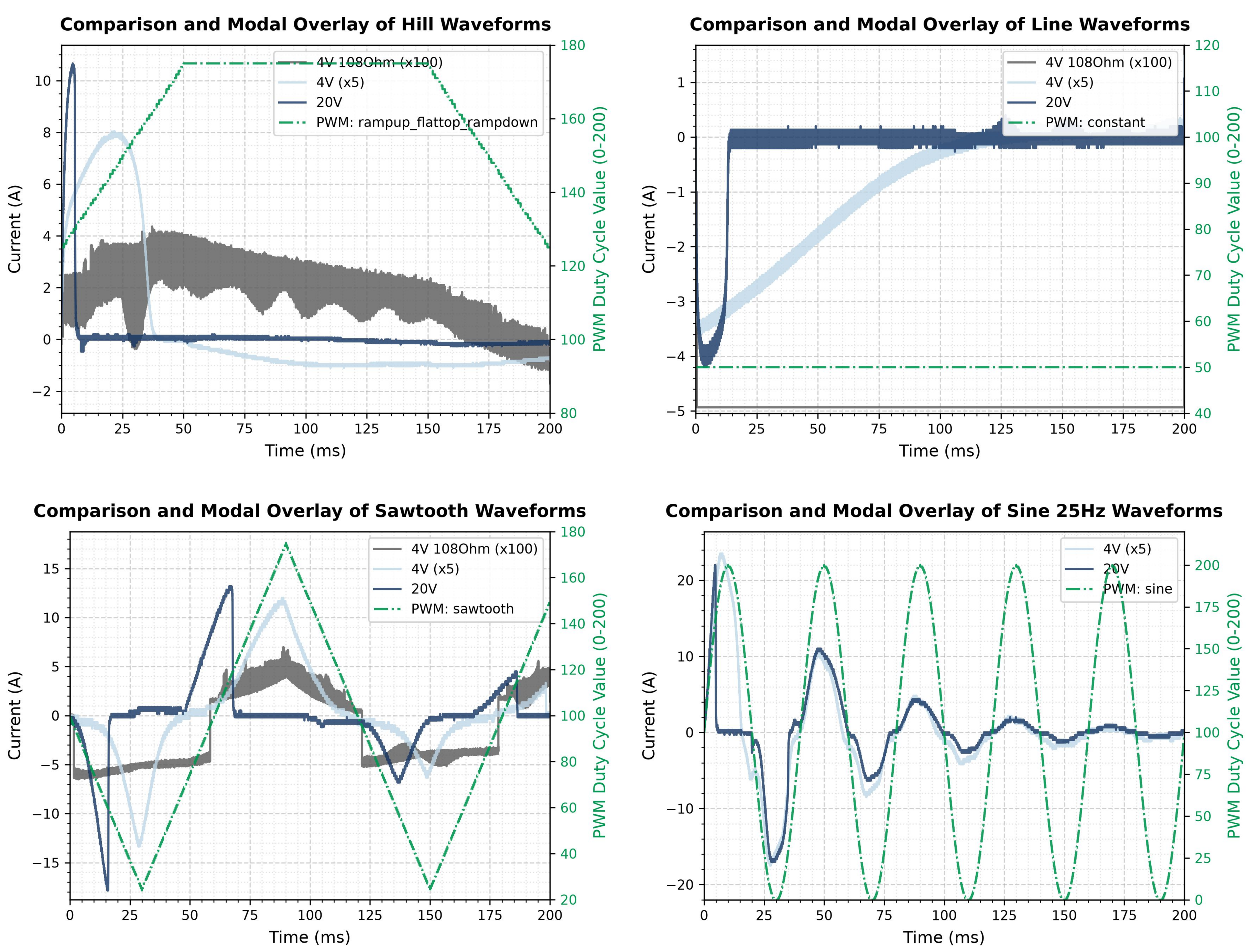
- A dummy inductive load was used to test H-bridge switching and current control at ±4V and ±20V.
- Tested diagnostics via a Pearson current probe and a shunt resistor, while monitoring voltage across the capacitor banks.



Voltage and current waveforms

Low Voltage Testing Results

- Waveforms acquired follow PWM inputs: scaling demonstrates consistency among duty-cycle mapping.
- Pearson's signal decays over constant current regions.



Coil current plots

Conclusions

- Fine-tuned, tested, and duplicated a Pi-to-Pico safe control chain on the SPA-controlled dummy coil.
- Built and tested an Ethernet data acquisition pipeline using PyVISA and tm_devices to capture, scale, and filter transient signals from the Tektronix oscilloscope.
- Future steps: scale the system coordinating multiple Picos and integrating a central network switch to allow simultaneous execution of multiple control and oscilloscope scripts.

Acknowledgement

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